

Estimation

1. Model Notations & Basics

- **ARMA(p, q):** $\phi(B)(Y_t - \mu) = \theta(B)Z_t$ where $Z_t \sim WN(0, \sigma^2)$.
- **Sample Moments:**
 - Mean: $\bar{Y} = \frac{1}{n} \sum_{t=1}^n Y_t$.
 - Variance: $C_0 = \frac{1}{n} \sum_{t=1}^n (Y_t - \bar{Y})^2$.
 - ACVF: $C_k = \frac{1}{n} \sum_{t=1}^{n-k} (Y_t - \bar{Y})(Y_{t+k} - \bar{Y})$.
 - ACF: $r_k = C_k / C_0$.

2. Method of Moment Estimators (MM)

Assume $\{Y_t\} \sim \text{ARMA}(p, q)$ is stationary with ACVF $\gamma(\cdot)$ and ACF $\rho(\cdot)$. $\gamma_0 = \sigma^2$

POPULATION QUANTITY	ESTIMATOR
μ	$\bar{Y}_n = \frac{1}{n} \sum_{t=1}^n Y_t$
$\gamma(k)$	$C_k = \frac{1}{n} \sum_{t=1}^{n-k} (Y_t - \bar{Y})(Y_{t+k} - \bar{Y})$
$\rho(k)$	$r_k = \frac{C_k}{C_0}$

3. Yule-Walker Estimators (YW) - AR Models Only

Matrix Form (The YW Equation):

$$\hat{\phi} = \begin{pmatrix} \hat{\phi}_1 \\ \vdots \\ \hat{\phi}_p \end{pmatrix} = \begin{pmatrix} 1 & r_1 & \dots & r_{p-1} \\ r_1 & 1 & \dots & r_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p-1} & r_{p-2} & \dots & 1 \end{pmatrix}^{-1} \begin{pmatrix} r_1 \\ \vdots \\ r_p \end{pmatrix}$$

然后把里面的对应的r替换成C

Variance Estimation: $\hat{\sigma}^2 = C_0 - \sum_{k=1}^p \hat{\phi}_k C_k$

4. Least Squares Estimators (LSE)

A. Unconditional Least Squares (ULS) - For AR(p)

对于 $Y_t \sim AR(p)$, $Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + Z_t \Rightarrow Y = X\phi + Z$

$$\mathbf{X} = \begin{pmatrix} Y_p & Y_{p-1} & \cdots & Y_1 \\ Y_{p+1} & Y_p & \cdots & Y_2 \\ \vdots & \vdots & \ddots & \vdots \\ Y_{n-1} & Y_{n-2} & \cdots & Y_{n-p} \end{pmatrix} \quad \text{and} \quad \mathbf{Y} = \begin{pmatrix} Y_{p+1} \\ Y_{p+2} \\ \vdots \\ Y_n \end{pmatrix}, \quad \Gamma_p = \begin{pmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(p-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(p-1) & \gamma(p-2) & \cdots & \gamma(0) \end{pmatrix}$$

Estimators: $\hat{\phi} = (X^T X)^{-1} X^T Y$. $\hat{\sigma}^2 = (Y - X\hat{\phi})^T (Y - X\hat{\phi}) / (n - 2p)$

Alternative Calculation: Minimize $S(\phi) = \sum_{t=p+1}^n (Y_t - \sum \phi_k Y_{t-k})^2$

Inference (Asymptotic Normality):

when $n \rightarrow +\infty$ $\sqrt{n}(\hat{\phi} - \phi) \rightarrow N_p(0, \sigma^2 \Gamma_p^{-1})$ where Γ_p is the covariance matrix of the process.

General C.I.: $\hat{\phi}_k \pm 2\sqrt{\hat{Var}(\hat{\phi}_k)}$ where $\hat{Var}$ comes from diagonal of $\hat{\sigma}^2 \hat{\Gamma}_p^{-1} / n$.

Also we have, for $\mathbf{w} = (w_1, \dots, w_p)^T \in \mathbb{R}^p$, as $n \rightarrow \infty$, $\sqrt{n}(\mathbf{w}^T \hat{\phi} - \mathbf{w}^T \phi) \xrightarrow{d} N(0, \sigma^2 \mathbf{w}^T \Gamma_p^{-1} \mathbf{w})$. It could be useful in testing a composite hypothesis, for example, $H_0 : \phi_1 = \phi_2$.

We can estimate Γ_p by $\hat{\Gamma}_p$ by replacing all $\gamma(k)$ by C_k .

B. Conditional Least Squares (CLS) - For MA(q) / ARMA(p,q)

Let $\{Y_t\} \sim \text{ARMA}(p, q)$ be **invertible**. Then

1. **(Initialization)** Assume $\tilde{Z}_s = Y_s = 0$ for all $s \leq 0$.
2. **(Sequential Estimation of Noise)** Let $\tilde{Z}_t = Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p} + \theta_1 \tilde{Z}_{t-1} + \cdots + \theta_p \tilde{Z}_{t-p}$.
3. **(Quantification of Error)** Define the sum of squared error as $S_*(\phi, \theta) = \sum_{t=1}^n \tilde{Z}_t^2$.

Then the **Conditional-Least Squares Estimators** are given by the minimizer of $S_*(\phi, \theta)$, i.e.

$$(\hat{\phi}, \hat{\theta}) = \arg \min_{\phi, \theta} S_*(\phi, \theta)$$

5. Maximum Likelihood Estimators (MLE)

1. 构建 Joint PDF

MLE 的核心是写出观测数据 $Y_1, Y_2, \dots, Y_n$ 的 **Joint PDF** $f(y_1, \dots, y_n; \theta)$ ，这在统计学中被称为 似然函数 $L(\theta)$ 。

由于时间序列数据不是独立的（ Y_t 依赖于过去的 Y_{t-1} ），我们不能简单地将概率相乘。我们主要通过两种方法来构建这个联合密度函数：

方法一：Iterative Conditioning

公式分解： $f(Y_1, \dots, Y_n) = f(Y_1) f(Y_2|Y_1) f(Y_3|Y_2, Y_1) \dots f(Y_n|Y_{n-1}, \dots, Y_1)$

以 AR(1) 为例 ($Y_t = \phi Y_{t-1} + Z_t$):

1. 条件部分 ($t \geq 2$): 给定 Y_{t-1} ，当前的 Y_t 服从正态分布，均值为 ϕY_{t-1} ，方差为 σ^2 ，即 $Y_t | Y_{t-1} \sim N(\phi Y_{t-1}, \sigma^2)$ 。

2. 初始部分 ($t = 1$): 为了保证精确性, 需要 Y_1 的边缘分布。对于平稳 AR(1), $Y_1 \sim N(0, \frac{\sigma^2}{1-\phi^2})$ 。

3. 最终似然函数: 将上述两部分相乘得到 $L(\phi, \sigma^2)$ 。

方法二: Multivariate Normal / Matrix Approach

这种方法不进行拆解, 而是将整个时间序列向量 $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ 视为一个整体, 服从多元正态分布。

- 公式: $L(\theta, \sigma^2) = (2\pi)^{-n/2} |\Sigma|^{-1/2} \exp \left\{ -\frac{1}{2} \mathbf{y}^T \Sigma^{-1} \mathbf{y} \right\}$
- 关键点: 这里的 Σ 是协方差矩阵, 其中元素 $\Sigma_{ij} = \gamma(|i - j|)$ ($n \times n$ 矩阵), 是参数 ϕ, θ, σ^2 的函数。MLE 就是要调整这些参数, 使得这个矩阵最能描述数据的相关结构。

2. 最大化似然函数

有了似然函数 $L(\theta)$ 后, 我们的任务是找到 $\hat{\theta}$ 让 L 最大。

1. 对数化 (**Log-Likelihood**): $l(\theta) = \ln L(\theta)$ 。AR(1): $l(\phi, \sigma^2) = \frac{1}{2} \ln(1 - \phi^2) - \frac{n}{2} \ln(\sigma^2) - \frac{S(\phi)}{2\sigma^2}$, 其中 $S(\phi) = \sum_{t=2}^n (Y_t - \phi Y_{t-1})^2 + (1 - \phi^2) Y_1^2$ 。

2. 数值优化 (**Numerical Optimization**): 对于 MA(q) 或 ARMA 模型, 通常没有闭式解, 需要用 Newton-Raphson、Scoring Method 等数值方法来最大化对数似然。

Comparison

Category 1: AR Models

1. ULS

Pros: Fast & Simple: It transforms into a standard linear regression problem. **Closed-Form Solution:** Has an exact analytical formula; no iteration required.

Cons: Information Loss: Treats the first p observations as fixed constants, ignoring their probability distribution. **Small Sample Bias:** Less accurate than MLE when sample size n is small.

2. MLE

Pros: Precision: Uses full information, including the distribution of initial values Y_1 . **Stationarity:** Naturally imposes stationarity constraints on parameters.

Cons: Computation: No closed-form solution; requires numerical optimization.

Category 2: MA & ARMA Models

1. CLS

Pros: Feasibility: Makes calculation possible by recursively estimating unobservable noise Z_t . **Efficiency:** Computationally faster than exact MLE; estimates are very close to MLE for large n .

Cons: Initialization Bias: Relies on the arbitrary assumption that initial errors $Z_0 = 0$, which causes bias in small samples. **Constraint:** Requires the model to be invertible.

2. MLE

Pros: Optimal: The most theoretically efficient estimator; utilizes the full covariance matrix Σ . **No**

Approximation: Does not need to assume $Z_0 = 0$.

Cons: Computationally Expensive: Calculation of the inverse covariance matrix Σ^{-1} or iterative conditioning is complex and slow.

Model Selection and Diagnostics

1. ACF & PACF

$PACF \phi_{kk}$ 就是转换为AR模型，然后看 Y_{t-k} 的系数。这两个参数能看的前提是都是Stationary的

MODEL	PLOT	PATTERN
MA(q)	ACF $\rho(k)$	Cut-off at lag q (zero for $k > q$)
	PACF ϕ_{kk}	Exponential Decay (MA(1)) (NA)
AR(p)	ACF $\rho(k)$	Exponential Decay (AR(1)) (NA)
	PACF ϕ_{kk}	Cut-off at lag p (zero for $k > p$)

2. Order Selection

$\hat{\beta} = (\hat{\phi}_1, \dots, \hat{\phi}_p, \hat{\theta}_1, \dots, \hat{\theta}_q)$ and $\hat{\sigma}^2$ as the MLE of the model given $Y_1, \dots, Y_n$.

$S_Y(\hat{\beta}) = \sum_{t=1}^n \hat{Z}_t^2$, where $\hat{Z}_t = Y_t - \hat{\phi}_1 Y_{t-1} - \dots - \hat{\phi}_p Y_{t-p} - \hat{\theta}_1 Z_{t-1} - \dots - \hat{\theta}_q Z_{t-q}$.

$L(\hat{\beta}, \hat{\sigma}^2) = (2\pi\hat{\sigma}^2)^{-n/2} \exp\{-S_Y(\hat{\beta})/(2\hat{\sigma}^2)\}$, as the likelihood function of ARMA(p, q) model.

$$\frac{S_Y(\hat{\beta})}{n} = \hat{\sigma}^2.$$

A. AIC (Akaike Information Criterion) $AIC = -2 \log L(\hat{\beta}, \hat{\sigma}^2) + 2(p + q + 1)$

Principle: AIC aims to estimate the **Expected Predictive Log-likelihood**. It strikes a balance between **model goodness-of-fit** (the likelihood function) and **model complexity** (the number of parameters $p + q + 1$).

Applicable Scenarios: Use when your primary goal is to achieve **accurate prediction**.

Drawback: AIC is **Not Consistent**; even with a large sample size n , it may select a model more complex than the true model. It tends to **overfit** with small sample sizes.

B. AICC (Corrected AIC) $AICC = -2 \log L(\hat{\beta}, \hat{\sigma}^2) + \frac{2(p+q+1)n}{n-p-q-2}$

Principle: AICC is a **small-sample correction** of AIC. When the sample size n is small, AIC tends to select models with too many parameters (overfitting). AICC introduces the correction factor $\frac{n}{n-p-q-2}$, which imposes a harsher penalty on the number of parameters when n is small.

Applicable Scenarios: You **must use AICC** to prevent overfitting when the sample size n is small. Since AICC converges to AIC when n is large, it is a generally more robust choice.

C. BIC (Bayesian Information Criterion)

$$BIC = (n - p - q) \log \left[\frac{n\hat{\sigma}^2}{n-p-1} \right] - n(1 + \log \sqrt{2\pi}) - (p + q) \log \left[\frac{\sum_{i=1}^n X_i^2 - n\hat{\sigma}^2}{p+q} \right].$$

Principle: Based on **Bayesian inference**, BIC aims to estimate the **Marginal Log-likelihood**. BIC assumes the existence of a "true model" and penalizes model complexity **more heavily** than AIC (the penalty term involves $\log n$).

Applicable Scenarios: BIC is a **Consistent** estimator. Use BIC when your goal is not purely prediction, but to find the **"true" underlying model structure** of the data (i.e., identifying the correct p and q) and the sample size n is sufficiently large.

D. FPE (Final Prediction Error, AR Model Only) $FPE = \left(\frac{n+p}{n-p}\right) \hat{\sigma}^2$

Principle: FPE is designed to estimate the **Mean Squared Error (MSE)** of the parameter estimators, specifically $E[(\hat{\phi} - \phi)'(\hat{\phi} - \phi)]$. It reflects the average predictive error variance when performing a one-step-ahead forecast using this model.

Applicable Scenarios: Only applicable to AR models. If you are fitting a pure AR model and wish to minimize the prediction error, you can use FPE. For large samples, the results selected by FPE are generally very similar to those selected by AIC (both lack consistency).

3. Model Diagnostics 检测Residuals, 看是否足够小, 并且可以看是否是WN

Step 1: Residual $\hat{Z}_t = Y_t - \hat{Y}_t$, 计算 $\hat{r}_Z(j) = \frac{\sum_{t=1}^{n-j} (\hat{Z}_t - \bar{Z})(\hat{Z}_{t+j} - \bar{Z})}{\sum_{t=1}^n (\hat{Z}_t - \bar{Z})^2}$

Step 2: Plot $\hat{r}_Z(j)$ 的图看是不是在 $(-2/\sqrt{n}, +2/\sqrt{n})$

Step 3: Apply the Ljung-Box test.

Ljung-Box:

Let $r_Z(j)$ be the sample ACF of $\{\hat{Z}_t\}$ and ρ_Z be the ACF of the true noise sequence.

$H_0 : \rho_Z(k) = 0$ whenever $|k| \leq h$ against $H_1 : \rho_Z(k) \neq 0$ for some $|k| \leq h$

The Ljung-Box Test is defined by $Q(h) = n(n+2) \sum_{j=1}^h \frac{r_{\hat{Z}}^2(j)}{n-j}$ and

$Q(h) \xrightarrow{d} \chi^2(h-p-q)$ under H_0 as $n \rightarrow \infty$.

Remark: 1. A common choice of h lies between **10 and 30**. 2. If $Q(h) \geq \chi_{h-p-q, 0.95}^2$, H_0 is **rejected** $\Rightarrow$ the **model is not a bad fit to the data**.

4. Four Stages of Model Building:

1. Pre-processing: (1) Remove trend and seasonal effect. (2) Check ACF of the residual to see if it is stationary. Otherwise, take differences to make it stationary

2. Model Estimation: (1) Draw ACF/PACF of the filtered data and get an initial ARMA model (2) Do estimation

3. Model Identification: Repeat step 2 for various ARMA models, choose an ARMA model by FPE/AIC/BIC

4. Model Checking: (1) Residual analysis (ts.plot/acf of $\hat{Z}_t$) (2) Ljung-Box test

Forecasting

1. Model Setup & Assumptions

- **ARIMA $\rightarrow$ ARMA** (差分)

若 $\{Y_t\} \sim ARIMA(p, d, q)$, 令 $X_t = \Delta^d Y_t$, 则 $\{X_t\} \sim ARMA(p, q)$ 。

- **X_t Assumption**

Causal: $X_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$ 且 $\sum |\psi_i| < \infty$ 。

Invertible: $Z_t = \sum_{i=0}^{\infty} \pi_i X_{t-i}$ 且 $\sum |\pi_i| < \infty$ 。

设 $Z_t = 0$ ($t \leq p$) , 对 $t = p+1, \dots, n$ 用 $Z_t = \sum_{k=1}^q \theta_k Z_{t-k} + (X_t - \sum_{k=1}^p \phi_k X_{t-k})$ 递推。

2. ARMA(p,q) 的 Box-Jenkins 预测

- 因果表示: $Y_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$, $Z_t \sim WN(0, \sigma^2)$ 。
- **h 步预测:** $\hat{Y}_{n+h} = E(Y_{n+h} | Y_1, \dots, Y_n) = \sum_{i=h}^{\infty} \psi_i Z_{n+h-i}$ (仅用已知 Z) 。

- 预测误差与方差:

$$e_n(h) = Y_{n+h} - \hat{Y}_{n+h} = \sum_{i=0}^{h-1} \psi_i Z_{n+h-i},$$

$$P_{n+h}^n = \text{Var}(e_n(h) \mid Y_1, \dots, Y_n) = \sigma^2 \sum_{i=0}^{h-1} \psi_i^2.$$
- 95% 区间: $\hat{Y}_{n+h} \pm 1.96 \sqrt{P_{n+h}^n}$.

3. ARIMA(p,d,q) 的预测 (反差分)

- 先预测差分序列: 对 $X_t = \Delta^d Y_t$ 用 ARMA 方法得 $\hat{X}_{n+1:n+h}$.
- 回到原序列 (以 $d=1$ 为例): $Y_{n+h} = Y_n + \sum_{j=1}^h X_{n+j}$, 因而 $\hat{Y}_{n+h} = Y_n + \sum_{j=1}^h \hat{X}_{n+j}$, 误差 $e_{n,Y}(h) = \sum_{j=1}^h e_{n,X}(j)$.
- 95%区间为 $\hat{Y}_{n+h} \pm 1.96 \sqrt{P_{n+h,Y}^n}$

Non-Stationary

1. Nonstationarity in Mean: TS vs. DS

- **Trend Stationary (TS):** $Y_t = \beta_0 + \beta_1 t + \nu_t$. Has deterministic trend. **Fix:** Detrending (Regression).
- **Difference Stationary (DS):** $Y_t = \beta_1 + Y_{t-1} + \nu_t$ (Random Walk with drift). Has stochastic trend. **Fix:** Differencing ($\Delta Y_t = Y_t - Y_{t-1}$).
- **Encompassing Model:** $Y_t = \beta_0 + \beta_1 t + v_t$; $v_t = \alpha v_{t-1} + Z_t$ 1代入2花间可得 $Y_t = \gamma_0 + \gamma_1 t + \alpha Y_{t-1} + Z_t$.
- If $\alpha = 1 \rightarrow$ **DS** (Unit Root, Non-stationary). If $|\alpha| < 1 \rightarrow$ **TS** (Mean Reverting, Stationary).

Unit Root Test (Dickey-Fuller Test) 查是不是trend stationary (但是只考 $Y_t = \alpha Y_{t-1} + Z_t$, 所以直接是 Stationary)

- **Hypothesis:** $H_0 : \alpha = 1 \Leftrightarrow \rho = 0$ vs. $H_1 : \alpha < 1$ (trend stationary).
- **Test Statistic:** AR coefficient $\hat{\alpha} = \frac{\sum Y_t Y_{t-1}}{\sum Y_{t-1}^2}$. Statistic $T = n(\hat{\alpha} - 1)$.
- **Distribution:** Follows Dickey-Fuller Distribution (FCLT):

$$n(\hat{\alpha} - 1) \xrightarrow{d} \frac{\int_0^1 W(t) dW(t)}{\int_0^1 W^2(t) dt}$$
 $W(t)$ 是 Brownian motion
- **Decision:** Reject H_0 if statistic < Critical Value or p-value $\leq 0.05 \Rightarrow$ Stationary

FCLT: $Y_k = \sum_{i=1}^k Z_i, \quad \frac{Y_{[nt]}}{\sqrt{n}} \xrightarrow{d} W(t).$

Unit root statistic: $\hat{\alpha} = \frac{\sum_{t=2}^n Y_t Y_{t-1}}{\sum_{t=2}^n Y_{t-1}^2}$

Denominator: $\sum_{t=2}^n Y_{t-1}^2 = n^2 \sum_{t=2}^n \left(\frac{Y_{t-1}}{\sqrt{n}} \right)^2 \frac{1}{n} \approx n^2 \int_0^1 W^2(t) dt$

Numerator: $\sum_{t=2}^n Y_t Y_{t-1} = \sum_{t=2}^n Y_{t-1}^2 + n \sum_{t=2}^n \frac{Y_{t-1}}{\sqrt{n}} \frac{Y_t - Y_{t-1}}{\sqrt{n}} \approx n^2 \int_0^1 W^2(t) dt + n \int_0^1 W(t) dW(t)$

2. Nonstationarity in Variance

Case A: Deterministic Change (Variance Stabilization) $Y_t = \mu_t + Z_t$

- **Situation:** Variance depends on mean, $Var(Y_t) = \sigma^2 h^2(\mu_t)$.
- **Goal:** Find transformation $g(Y_t)$ such that $Var(g(Y_t)) \approx \text{Constant}$.
- **Formula:**
$$g(Y) \approx g(\mu) + g'(\mu)(Y - \mu) \Rightarrow Var(g(Y)) \approx [g'(\mu)]^2 h^2(\mu) \sigma^2.$$

To make variance constant, set $g'(\mu) = \frac{k}{h(\mu)}$ and $g(x) = k \int \frac{1}{h(\mu)} d\mu$

- **Box-Cox Transformation Family ($g_\lambda(y)$):**

$$g_\lambda(y) = \begin{cases} \frac{y^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \ln(y) & \lambda = 0 \end{cases}$$

Case B: Stochastic Change (Heteroskedasticity)

- **Stylized Facts of Financial Returns ($X_t = \Delta \ln P_t$):**
 - a. **Heavy-tailed:** Kurtosis > 3 (Not Normal).
 - b. **Uncorrelated X_t :** ACF of returns ≈ 0 (White Noise).
 - c. **Correlated X_t^2 :** ACF of squared returns $\neq 0$ (**Volatility Clustering**).

3. ARCH(q) Model

- **Definition:** $X_t = \sigma_t \epsilon_t$, where $\epsilon_t \sim i.i.d. N(0, 1)$.
$$\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \dots + \alpha_q X_{t-q}^2 = var(X_t | \mathcal{F}_{t-1})$$
- **Constraints:** $\alpha_i \geq 0$ (Positivity), $\sum_{i=1}^q \alpha_i < 1$ (Stationarity).

ARCH(q) $\Leftrightarrow$ AR(q) for X_t^2

The white-noise sequence is given by $v_t = X_t^2 - \sigma_t^2 = \sigma_t^2(\epsilon_t^2 - 1)$. Then $\{X_t^2\} \sim \text{ARMA}(q, 0)$ which is an AR(q) process, as

$$X_t^2 = \sigma_t^2 + (X_t^2 - \sigma_t^2) \text{ Noticed that } \sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i X_{t-i}^2.$$

$$X_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i X_{t-i}^2 + v_t \text{ where } E(v_t) = 0 \text{ and } \text{Cov}(v_t, v_{t-h}) = 0 \text{ for } h \neq 0.$$

Properties (ARCH(1)):

1. Unconditional Mean: $E[X_t] = 0$.
2. Unconditional Variance: $\text{Var}(X_t) = E[X_t^2] = \frac{\alpha_0}{1-\alpha_1}$ (Condition: $\alpha_1 < 1$).
3. Kurtosis (4th Moment): $E(X_t^4) = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)}$. Exists only if $3\alpha_1^2 < 1$.
4. ACF of X_t^2 : $\text{Corr}(X_t^2, X_{t+h}^2) = \alpha_1^{|h|}$ (Decays like AR process).

4. GARCH(p, q) Model

- **Definition:** Adds lagged variance terms.
$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i X_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$
- **Stationarity 成立条件:** $\sum \alpha_i + \sum \beta_j < 1$.

Comparison: Unconditional vs. Conditional Moments (Stationary GARCH)

QUANTITY	UNCONDITIONAL (LONG-RUN)	CONDITIONAL ON $\mathcal{F}_{t-1}$ (SHORT-RUN)
Mean	$E(X_t) = 0$	$E(X_t \mathcal{F}_{t-1}) = 0$
Variance	$Var(X_t) = \frac{\alpha_0}{1-(\sum_q \alpha_i + \sum_p \beta_j)}$	$Var(X_t \mathcal{F}_{t-1}) = \sigma_t^2$
Covariance	$Cov(X_t, X_{t+h}) = 0$	$Cov(X_t, X_{t+h} \mathcal{F}_{t-1}) = 0$

I-GARCH (Integrated Non-Stationary):

If $\alpha_1 + \beta_1 = 1$. Non-stationary.

Forecasting variance: $E[\sigma_{t+j}^2|\mathcal{F}_{t-1}] = j\alpha_0 + \sigma_t^2$ (Linear growth).

KEY DERIVATION: GARCH(p,q) $\Leftrightarrow$ ARMA(m, p) for X_t^2

Conditions: 1. 是Stationary的 2. $E(\sigma_t^4)$ exists and is constant over-time

The white-noise sequence is given by $v_t = X_t^2 - \sigma_t^2 = \sigma_t^2(\epsilon_t^2 - 1)$. Then $\{X_t^2\} \sim \text{ARMA}(m, p)$ as

$$X_t^2 = \sigma_t^2 + (X_t^2 - \sigma_t^2) = \alpha_0 + \sum_{j=1}^q \alpha_j X_{t-j}^2 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2 + (X_t^2 - \sigma_t^2)$$

$$= \alpha_0 + \sum_{j=1}^m (\alpha_j + \beta_j) X_{t-j}^2 - \sum_{i=1}^p \beta_i (X_{t-i}^2 - \sigma_{t-i}^2) + v_t = \alpha_0 + \sum_{j=1}^m (\alpha_j + \beta_j) X_{t-j}^2 + v_t - \sum_{i=1}^p \beta_i v_{t-i}$$

where $\alpha_i := 0$ and $\beta_j := 0$ for $i \geq q, j \geq p$.

Compare In an ARCH(q) model the conditional variance σ_t^2 depends **only** on past shocks X_{t-i}^2 , so to capture long-lasting volatility clustering we often need a **large q**.

GARCH adds lagged variances σ_{t-j}^2 on top of past shocks, so the variance “remembers” the distant past through its own lags.

5. Estimation & Testing Flow

异方差 (Heteroscedasticity) 同方差 (Homoscedasticity)

A. Detecting Heteroskedasticity (LM Test)

H_0 : No ARCH effect $\alpha_1 = \dots = \alpha_p = 0$ H_1 : ARCH effect exists

Run auxiliary regression of squared residuals: $X_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \dots + \alpha_q X_{t-q}^2$

Statistic: $T = n \times R^2 \rightarrow \chi^2(q)$.

Rule: If $T > \chi_{0.95,p}^2$ (or p-value < 0.05), **Reject** H_0

B. Estimation (MLE)

$$f(x_t | \mathcal{F}_{t-1}) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{1}{2\sigma_t^2} x_t^2\right)$$

Method: Use Maximum Likelihood (MLE), **NOT OLS** (because variance changes).

Log-Likelihood Function (to maximize):

$$l(\theta) = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \sum_{t=1}^n \left(\ln(\sigma_t^2) + \frac{X_t^2}{\sigma_t^2} \right)$$

(Must compute σ_t^2 recursively from $t = 1$ to n).

C. Model Selection:

Use **AIC** / **BIC**.

$$AIC = -2\log L + 2(p + q + 1) \quad BIC = -2\log L + (p + q + 1)\log n$$

Note: PACF of X_t^2 works for ARCH order identification, but fails for GARCH.

D. Diagnostics (Goodness of Fit)

Object: Standardized Squared Residuals: $\hat{\epsilon}_t^2 = \frac{X_t^2}{\hat{\sigma}_t^2}$.

Test: Ljung-Box Test on $\hat{\epsilon}_t^2$. H_0 : no autocorrelation in ϵ_t^2 up to lag h .

Statistic: $Q(h) = n(n+2) \sum_{j=1}^h \frac{r^2(j)}{n-j} \rightarrow \chi^2(h-p-q-1)$.

Goal: Fail to reject H_0 (High p-value > 0.05) $\rightarrow$ No correlation left $\rightarrow$ Good Model.